

Risk-Return Analysis of Dynamic Investment Strategies*

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Abstract

The investment management industry has developed such a wide range of trading strategies, that many investors feel lost when they have to choose the investment style that meets their requirements. Comparing these on a like-for-like basis is a difficult task about which much has been written. The scope of this paper is restricted to strategies investing in a single asset, and which are driven by the price of this asset. We show how those strategies can be fully characterised by two components: an option profile and some trading impact. The option profile depends solely on the final asset value, whereas trading impact is driven by the realised volatility. From this analysis, most of these investment strategies can be categorised in one of three families: directional, contrarian and trend-following. While directional strategies exhibit the same kind of behaviour as the underlying, contrarian and trend-following strategies exhibit asymmetric return distributions. Those asymmetric behaviours can be misleading at first sight, as a seemingly stable strategy may hide large potential losses.

Keywords: Dynamic strategies, option payoff, asymmetric returns, trend-following strategy, contrarian strategy, volatility.

JEL classification: G11, G17, C63.

1 Introduction

We are unable to list here the immense variety of “investment styles” that are used in the financial industry to generate returns. Value investors, growth investors, trend followers, global macro analysts, long-short equity managers, special situations specialists or volatility arbitragers, to name but a few, all rely on different and sometimes opposing views of how markets work. However, within each style, some people succeed and some do not, preserving the mystery of what are the determining factors of successful investment strategies. It is very likely that none of these strategies is able to predominate sufficiently to eliminate the others, as market forces would disable this very strategy in due course.

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As heterogeneous as those strategies might be, there is a need for final investors to compare them in order to understand which are the determining factors in their performance, how they compare and whether it makes sense to pay fees to a professional investment manager.

Some metrics such as Sharpe ratio, factor analysis and specialised index benchmarking have become widely accepted tools for analysing investment performance in the mutual fund universe. However, this multi-factor analysis fails to account for more complex strategies such as the ones used by hedge funds. Those funds use dynamic strategies that generate returns that are difficult to link to the behaviour of standard factors such as the main equity indexes. Imagine a stylised situation in which one has to compare two investment strategies. The first would systematically sell short puts on the S&P index while investing the proceeds in short-term bonds. The second would consist in investing eighty percent of its assets in short-term bonds and using the remaining cash to invest in quarterly call options on the S&P index. Which is the better strategy? How can they be compared? It is quite clear that a Sharpe ratio or linear factor analysis would not provide enough information to assess their quality.

Some attempts have been made to provide answers to those questions, some of which we will now review. Addressing non-linearities in returns both in pricing models and in fund performance is not a new topic. To quote only a few, Harvey and Siddique (2000) propose a factor model incorporating not only the returns but the square of the returns to explain hedge fund returns. Elsewhere, Agarwal and Naik (2004) propose a similar approach for analysing equity-oriented hedge fund performance. However, instead of using the square of returns, they create synthetic factors that mimic the performance of call options, introducing different kinds of non-linearities. Fung and Hsieh (2001) focus on trend-following strategies. Roughly, trend-followers invest in proportion to the past performance of a specific market, whether it is positive or negative. As a result, those strategies resemble the delta of a straddle option in qualitative terms. Fung and Hsieh (2001) are thus testing the hypothesis that the performance of trend-following strategies can reliably be compared to the simulated performance of a strategy rolling lookback straddles on the MSCI World index. In the professional world, many analysts classify strategies depending on whether they are convergent or divergent, depending on their tendency to go against the market or to follow it (see Chung *et al.* (2004), for instance).

The common feature of these approaches is that the design of the non-linear relevant factor is mostly based on qualitative considerations and is used to provide econometric tests of those assumptions. In this paper, we follow a slightly different route. Instead of trying to analyse real-life hedge fund returns, we aim to constructively identify the exact payoff generated by popular dynamic strategies.

The main idea for identifying this payoff is borrowed from option pricing literature. It is well known that the strategy for obtaining a call option payoff consists in investing its delta in the market on a day-to-day basis, this delta being the derivative of the price of the call. In so doing, one obtains the payoff of a call, less the so-called gamma costs. Imagine now that a portfolio strategy can be defined as a function of a given underlying asset price. It is very likely that the payoff generated by such a strategy is the primitive of this strategy exposure, plus or minus some trading impact. This covers those situations in which a manager can decide how much to invest merely by looking at the price and some fixed parameters. We show in this paper that many popular strategies fit this description.

This paper is thus structured as follows. In the second section, we show how simple strategy performances can be broken down into an option profile and some trading impact. Section 3 is dedicated to studying the properties of simple directional strategies consisting in holding a constant fraction of one's wealth in a given asset. In section 4, we develop an analysis of two popular contrarian strategies: the "return to average" strategy and the averaging down strategy and we list their common features. Lastly, section 5 is devoted to trend-following strategies, where an original result illustrates their typical convex behaviour.

2 Breaking investment strategies down into an option profile and some trading impact

First of all, let us emphasise that the aim of this white paper is not to insist on the mathematics of the results but on the financial messages that are obtained. As a result we often omit mathematical aspects such as filtrations, continuity of functions or detailed properties of processes that would be necessary for a rigorous presentation. We hope that what has been lost in terms of accuracy and rigour will be offset by the gain in simplicity and legibility.

2.1 Model and results

Consider a simplified situation in which an asset can be traded at each date t at price S_t , without any friction of any kind. S_t is supposed to be governed by an ordinary diffusion model:

$$\begin{aligned} dS_t &= S_t \times (\mu_t dt + \sigma_t dW_t) \\ S_0 &= s \end{aligned}$$

An investor is running an investment strategy consisting in holding a number $f(S_t)$ of securities at any time. This strategy is supposed to depend only on the price itself and some parameters, but not on time or on other state variables. For the sake of simplicity, it is assumed that interest rates are zero¹. The wealth of the investor at each date t is denoted X_t . On a day-to-day basis, or between t and $t + dt$, variation in wealth is written as:

$$dX_t = f(S_t) dS_t \tag{1}$$

If S_t was deterministic and non-stochastic, it would be clear that:

$$\begin{aligned} X_T - X_0 &= \int_{S_0}^{S_T} f(S_t) dS_t \\ &= F(S_T) - F(S_0) \end{aligned}$$

where:

$$F(S) \equiv \int_a^S f(x) dx,$$

whatever the a chosen. If it is not, Ito's lemma applied to the function F yields the important property we want to emphasise.

¹To recover results in situation with interest rates, S_t and X_t will have to be replaced respectively by $S_t e^{-rt}$ and $X_t e^{-rt}$ in the different equations.

Proposition 1 *Any portfolio strategy of the type (1) described above can be broken down into an option profile plus some trading impact as follows:*

$$X_T - X_0 = \underbrace{F(S_T) - F(S_0)}_{\text{option profile}} + \underbrace{-\frac{1}{2} \int_0^T f'(S_t) S_t^2 \sigma_t^2 dt}_{\text{trading impact}} \quad (2)$$

While simple, the above proposition yields some interesting qualitative properties. Regarding model assumptions, it is worth noticing the robustness of this property, which can be obtained whenever Ito's lemma can be used with continuous price processes. This covers a wide variety of models and does not rely on special probabilistic assumptions. These include Black-Scholes, of course, but also local and stochastic volatility models. On the other hand, the option profile obtained is European. If the strategy is richer in terms of variable states, the option profile will be a function of those different states.

Let us now elaborate on these two terms. Trading impact depends on the variation in the number of holdings in relation to market fluctuations. If the strategy involves buying when the market goes up ($f' > 0$) then the trading impact will necessarily be negative, illustrating the "buy high sell low" curse of trend followers. Conversely, strategies that play against the market will always have positive trading impact, benefiting from the opposite effect. This trading impact increases with the volatility.

Interestingly, the sign of trading impact is directly related to the convexity of the option profile. Positive trading impact is necessarily associated with concave profile. This is no surprise to those familiar with option hedging, where it is well known that hedging a convex profile will generate positive gamma gains, which explains the difference between option prices and their intrinsic value.

All strategies with positive trading impact share similar characteristics in terms of winning probability. When trading impact is positive, using Proposition 1 leads to:

$$\Pr\{X_T \geq X_0\} > \Pr\{F(S_T) \geq F(S_0)\}$$

In a situation where F is non-decreasing, we get:

$$\Pr\{X_T \geq X_0\} > \Pr\{S_T \geq S_0\}$$

The probability of showing a profit is therefore higher than the probability of the underlying asset going up. On a weekly basis, most financial assets have a near 50/50 probability of going up or down, which means that those strategies have more chance of showing a profit than a loss. The higher the trading impact or the more concave the option profile, the higher the probability of showing a profit. This effect is offset by higher potential losses. A concave profile will therefore always exhibit negative skewness. Using a plain realised Sharpe ratio to assess future fund performance is very likely to be flawed, as it would be inflated artificially by the frequent positive gains. Following this analysis, rather than indicating a good portfolio manager, frequent positive gains may be symptomatic of strategies with high possible losses. The reverse holds for strategies with negative trading impact.

This breakdown may be worth bearing in mind from a qualitative point of view. In some cases, it might be tempting to focus on one term and to neglect the other, but in general they are of equal importance. A typical example of such bias is the stop-loss overlay, which is commonly used to protect against losses in a portfolio.

2.2 Trading impact associated with the stop-loss overlay

Imagine a situation where an investor runs an investment strategy which value is denoted S_t . In order to limit its losses, this investor wants to add a separate overlay to the strategy, which would consist in doing nothing while the strategy is above a certain threshold level S_{stop} , and going short when it is below. In the following example, we will consider the initial strategy as an underlying asset, and focus on the stop-loss overlay analysis. This overlay is purely price dependent and should satisfy the assumptions of proposition 1. The corresponding function f , representing the number of risky asset shares in the overlay portfolio, would be:

$$f(S_t) = \begin{cases} 0 & \text{when } S_t > S_{\text{stop}} \\ -1 & \text{when } S_t \leq S_{\text{stop}} \end{cases}$$

Along the lines of Proposition 1, the option profile is a put option of strike S_{stop} . Trading impact is more difficult to assess since f cannot be differentiated in a traditional manner. Technically, one should use another version of Ito's formula, sometimes referred as Tanaka's formula. Interested readers can refer to Carr and Jarrow (1990) for a detailed analysis of that strategy. Qualitatively, f can somehow be differentiated and its differential is equal to zero everywhere except at S_{stop} , where it has an infinite positive value. Hence, trading impact will necessarily be negative, proportional to the volatility and to the time spent by the underlying asset around S_{stop} . In a risk-neutral world, the average trading impact of this stop-loss strategy is equal to the cost of the put. Interestingly, the stop-loss strategy which could appear to be a free lunch as compared to buying a put option, generates some trading impact, which is equal in average to the price of the put itself.

To confirm this effect, this strategy is simulated in Figure 1, with a stop loss level S_{stop} equal to 90% of the original price. The resulting wealth is well below the asset price. Indeed, this policy has a very strong trading impact around the strike level S_{stop} . Each time this level is crossed, the strategy suffers from significant trading costs (see Figure 2 for a description). These costs cause the wealth level to deviate from the target profile. Each time the stop

Figure 1: Stop loss/start gain strategy trajectory

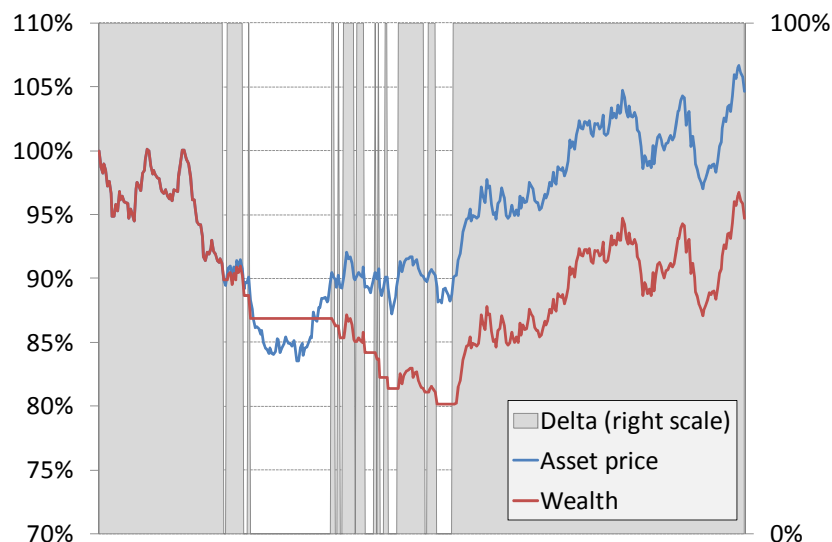
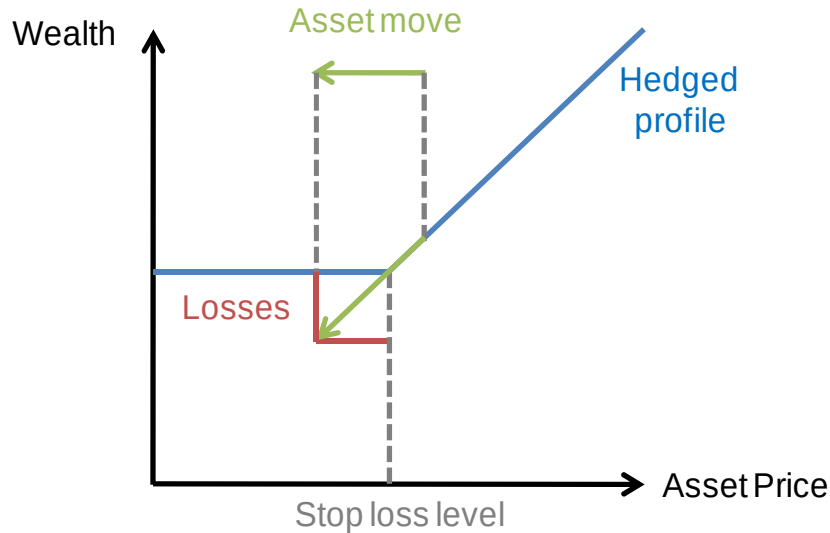


Figure 2: Trading impact appearance when crossing the strike



loss level is crossed, the loss with respect to the target profile is proportional to the size of the price movement. Total trading cost is therefore proportional to the volatility and to the number of times the asset price crosses the strike S_{stop} . In the following, we calculate the average trading cost, supposing that the asset price has a trend equal to the risk-free rate. In that case, the average trading cost is exactly equal to the price of the put option.

2.2.1 A tree-based approach

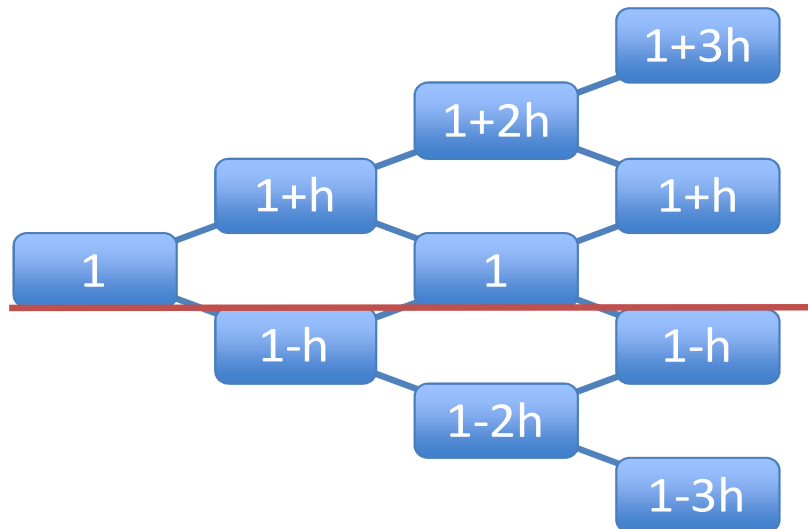
One might wonder whether this is an artefact of continuous time or not. To answer this legitimate question, we provide a discrete time analysis and show how a similar result can be obtained. We use a discrete tree-based approach to estimate trading costs, encountered each time the asset price crosses the stop-loss barrier. In this tree model, the asset price can increase or decrease by $\pm h$ at every time step. Each time step typically represents a business day. Thus the typical time step is $\delta t = 1/260$ years and the asset price variation size should be $h = \sigma\sqrt{\delta t}$ to obtain an annualised price volatility equal to σ . We obtain the tree-based representation of Figure 3, where the stop loss level $S_{\text{stop}} = 1 - \frac{h}{2}$ is represented by the red line, just below the initial price. Each time the price crosses this red line (from above or below), the investor loses $\frac{h}{2}$ with respect to the target payoff $\max(S_t, S_{\text{stop}})$. Therefore, the trading costs, i.e. the losses with respect to the target payoff, will be $\frac{h}{2}$ multiplied by the number of times that the level S_{stop} is crossed. The average number of times the barrier is crossed behaves like $\sqrt{\frac{2}{\pi} \frac{T}{\delta t}}$, where T is the total investment period, and δt is the time step². Thus the average trading cost is given by:

$$L \approx \frac{h}{2} \sqrt{\frac{2}{\pi} \frac{T}{\delta t}} \rightarrow \sigma \sqrt{\frac{T}{2\pi}}$$

This is exactly the price of the put option, in a Gaussian framework. This equality between the average trading cost and the price of the put option holds in any risk neutral model.

²This is the central limit probability for the asset value to quote at that value.

Figure 3: Tree based representation of the asset price



Compared to the protection based on the put option, the stop loss/start gain strategy has approximately the same average cost. But this cost is very uncertain, while the cost of the put option is known at inception. Even if the price of a put option is too high compared to the expected future volatility, a hedging policy can be applied to replicate this option (with the Black-Scholes formula). In this case, the cost of protection will depend on the realised volatility alone, irrespective of the number of times the strike is crossed.

A profit-taking strategy is the exact opposite of the stop loss strategy. Diametrically opposite results thus apply. As for the stop loss, investor can choose between selling a call option and a profit-taking strategy. The call option seller abandons returns above S_{stop} in exchange for receiving a fixed premium P . The definitive profit-taking strategy also abandons returns above the same threshold, but is not exposed to market drawdowns once the profits have been locked in. The profit-taking strategy that re-invests in the asset when the price is below the threshold abandons high asset performance, but receives trading gains each time the strike is crossed.

3 Directional strategies: balanced and leveraged funds

A common way to capture the risk premia yielded by equity markets consists in running investment strategies that invest a stable proportion of one's assets in risky markets. Merton (1971) shows that this strategy is indeed optimal for investors having a logarithmic utility and constant assumptions on expected returns and risks. Those strategies are sometimes referred to as constant-mix strategies. Retail networks are used to distribute products tagged as conservative, balanced or aggressive. They often implement constant-mix type of strategies, typically investing 20%, 50% or 80% of their wealth in risky markets. The popular 130/30 leveraged strategies are another example of constant-mix. Leveraged strategies whereby an investor maintains a constant leverage of 2 or 3 on some asset also belong to the constant-mix category. Constant proportion portfolio insurance (CPPI) strategies are a combination of constant mix strategies plus some zero-coupon bond. Eventually, a strategy consisting in

being short the market is a constant mix strategy with a leverage equal to -1 .

A widespread proxy to understand the behaviour of constant-mix is the equivalent buy and hold strategy. We will show in the following that, even if satisfactory for a short-term horizon, this approximation may turn to be misleading over an horizon of more than a few months.

3.1 Option profile and trading impact

For the sake of simplicity, we shall focus on a single asset case, similar to the one described in section 2. The constant proportion of wealth invested in the risky asset, exposure, is denoted by e . Locally, the relative variation of wealth is proportional to the return of the risky asset:

$$\frac{dX_t}{X_t} = e \frac{dS_t}{S_t} \quad (3)$$

where $\frac{dX_t}{X_t}$ is the return of the strategy between t and $t + dt$ and $\frac{dS_t}{S_t}$ is the return of the risky asset. This equation is slightly different from equation (1) as it involves exposure rather than number of shares. However, we can follow exactly the same route, and by applying Ito's lemma to $\ln X_t$ we get a similar proposition.

Proposition 2 *In the case of constant exposure e , the portfolio strategy can be broken down into an option profile multiplied by the exponential of the trading impact in the following way:*

$$X_T = X_0 \underbrace{\left(\frac{S_T}{S_0}\right)^e}_{\text{option profile}} \exp \left(\underbrace{\frac{1}{2} (e - e^2) \int_0^T \sigma_t^2 dt}_{\text{trading impact}} \right) \quad (4)$$

where $\int_0^T \sigma_t^2 dt$ is the cumulated variance of the risky asset between times 0 and T . If volatility is constant over time, this quantity is equal to $\sigma^2 T$.

As in Proposition 1, the wealth can be split at a certain date into an option profile and some trading impact. However, here the two terms are multiplied rather than added together. The option profile is a power option, whose power is exposure e , while the trading impact depends on realised volatility alone. Since both terms are positive, constant-mix strategies always ensure positive wealth in a market that trades continuously without any gap.

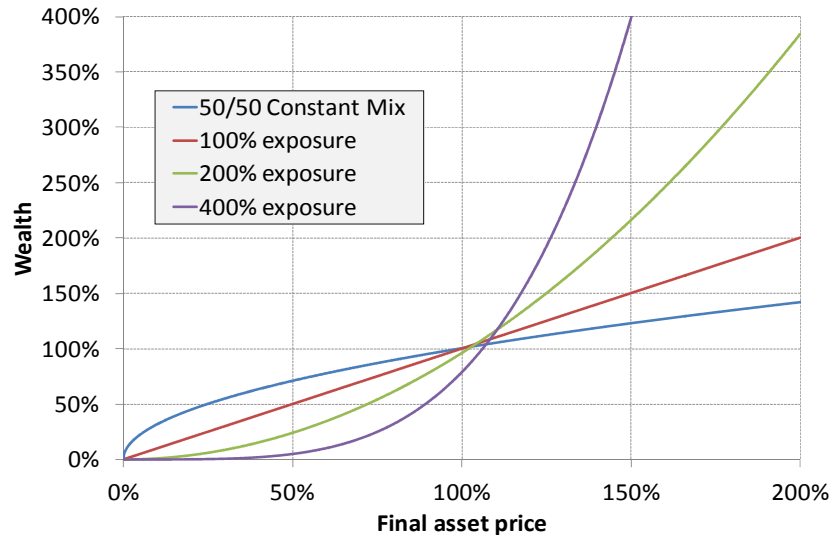
The $e - e^2$ term is a variation of holdings as in Proposition 1. For confirmation of this, let us consider the discrete case where a fraction of the wealth, e , is invested in a risky asset. Let f denote the number of securities held in the portfolio. Initially $f_t = \frac{eX_t}{S_t}$. If the risky asset moves by $x\%$ then we have:

$$\begin{aligned} S_{t+\delta t} &= S_t \times (1 + x) \\ X_{t+\delta t} &= X_t \times (1 + ex) \\ \Delta f &= e \frac{X_{t+\delta t}}{S_{t+\delta t}} - e \frac{X_t}{S_t} \end{aligned}$$

The variation of holdings can then be computed as

$$\frac{\Delta f}{\Delta S} = X_t \times \frac{e(e-1)}{S_t^2}$$

Figure 4: One year option profile adjusted from trading impact, with 20% volatility



The trading impact in equation (4) appears to be of the same kind as in equation (1) of Proposition 1 and related to the variation of number of holdings with respect to the variation of the risky asset. The option profile is not linear with respect to the asset price, except in the obvious cases of a delta one product ($e = 1$) or a portfolio fully invested in cash ($e = 0$).

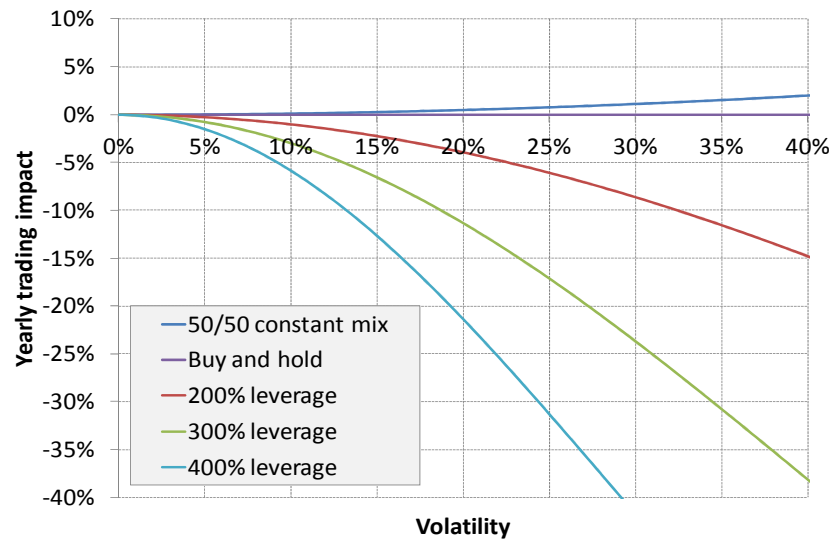
In the case of balanced funds, with positive exposure and no leverage, the option profile is concave and trading impact is positive (see Figure 4). In terms of indexation to the underlying market, the strategy is less indexed to the risky asset when its value is high, and more indexed to this asset when its value is low. On the other hand, leveraged strategies ($e > 100\%$) and short selling strategies ($e < 0$) exhibit a convex profile. Those profiles offer potentially very high returns at the cost of more frequent losses. Trading impact increases rapidly as e grows (see Figure 5). For example, the influence of volatility is 3 times larger³ for $e = 3$ than for $e = 2$. Figure 4 describes the effect of those strategies on final wealth, taking both target payoff and trading impact into account for 20% volatility and a 5-year horizon.

3.2 Predictions compared to actual backtests

All these formulas are derived from continuous time mathematical models, but they are quite accurate in practical situations. We compare backtested results of constant mix strategies combining the DJ Eurostoxx 50 index and cash. For each simulation, constant mix strategies start with an initial value equal to 1. Then, the value of the strategy after one year is compared to the relative value of the Eurostoxx 50 with respect to the starting date. Simulations are started on each business day between January 1987 and December 2010. Figures 6 and 7 plot the values of each backtest with respect to the relative value of the Eurostoxx 50. These values are compared to the prediction obtained with formula (4), where the volatility parameter is set equal to 20%. Figure 6 illustrates the balanced strategy (50% invested

³The expression $\frac{1}{2}(e - e^2)$ is equal to -3 when $e = 3$, and is equal to -1 if $e = 2$, while the term $\sigma^2 T$ remains unchanged for both exposures.

Figure 5: Annualised trading impact as a proportion of the initial wealth



in Eurostoxx and 50% invested in cash), while Figure 7 illustrates the 4 times leveraged Eurostoxx 50 strategy.

The cash rate is still taken to be 0. As shown in Figure 6, the prediction is very accurate for the 50/50 constant mix strategy. On the other hand, the 4 times leveraged strategy has a larger prediction error. This is because the prediction is computed with fixed volatility of 20%. The effective 1-year volatility of the Eurostoxx can be very different from this value. When $e = 50\%$, sensitivity to the realised variance is very low and equal to 12.5% of the

Figure 6: 50/50 constant mix strategy on the Eurostoxx 50

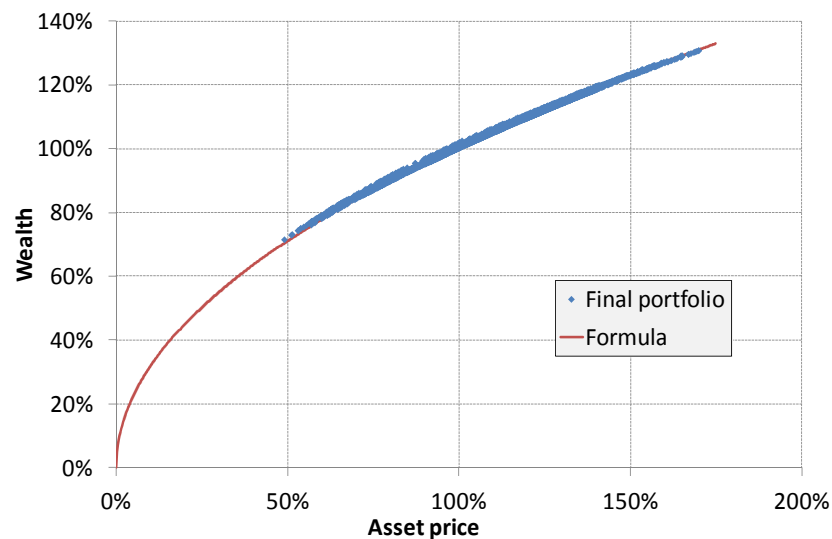
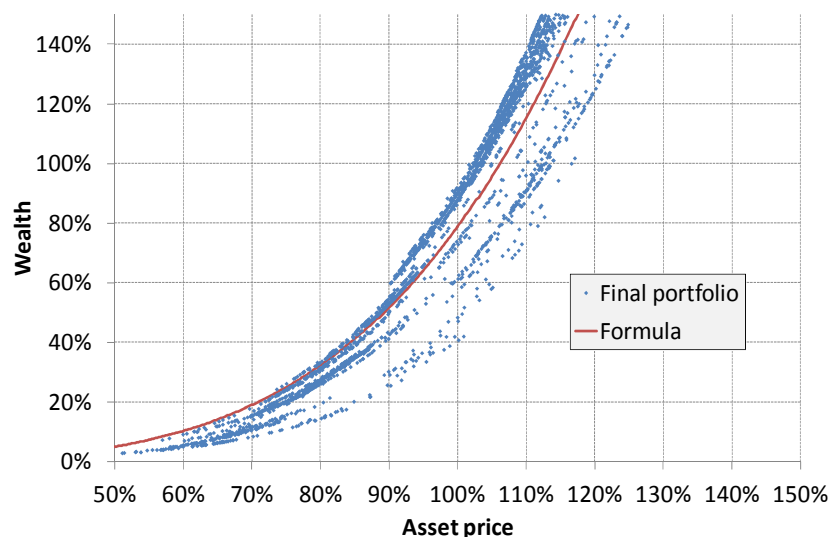


Figure 7: 4 times leveraged strategy on the Eurostoxx 50



variance⁴.

It is much higher for leveraged strategies. When $e = 4$ for instance, it is equal to -600% of the variance. The final value of the leveraged strategy thus depends heavily on the realised variance. This explains the differences with respect to the 20% volatility formula. As bullish markets are generally less volatile than bearish markets, backtested portfolios are generally higher than the formula for high returns of the Eurostoxx 50 index (right side of Figure 7), and lower than the formula for most negative Eurostoxx returns (left side of Figure 7).

4 Contrarian strategies

Let us now focus on contrarian strategies. From Proposition 1, we expect that, going against the market, those strategies will have a tendency to exhibit frequent small gains and less frequent large losses. In particular, this section is devoted to the study of two popular strategies: the mean-reversion strategy and the averaging down strategy.

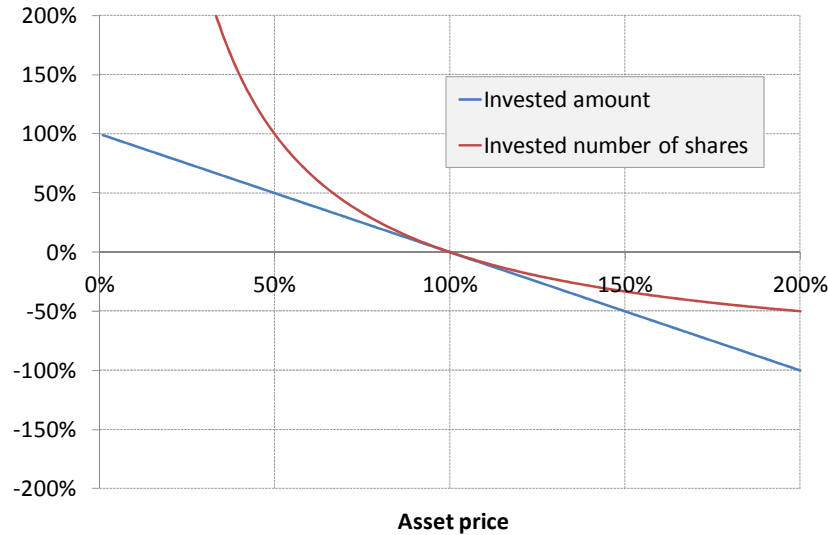
4.1 Mean-reversion strategies

4.1.1 Strategy definition

In some situations, investors state that an underlying should quote close to a price denoted S_{target} in the sequel. S_{target} can be obtained as the fundamental value of a stock obtained using financial analysis. It can also be a kind of average value around which an asset is supposed to exhibit some mean-reversion behaviour. Mean-reversion rationales are frequent in financial markets, as certain ratios are supposed to remain within a certain absolute range, outside which the situation is deemed abnormal. Price/earnings, volatility levels, spreads between stocks or indexes among others are indicators that are commonly used as a basis for mean-reversion analysis.

⁴Sensitivity to the cumulated variance $\sigma^2 T$ is equal to $\frac{1}{2}(e - e^2)$. This expression is equal to 12.5% when $e = 50\%$.

Figure 8: Mean-reverting investment policy



All these strategies can be summarised in a simple guideline: buy the asset when its price is below its target S_{target} , and sell it when the price is above. The simplest corresponding rule consists in holding an amount proportional to the distance between the price and its target level, S_{target} . In the same framework as Proposition 1 this can be written as:

$$f(S_t) = \frac{m(S_{target} - S_t)}{S_t} \quad (5)$$

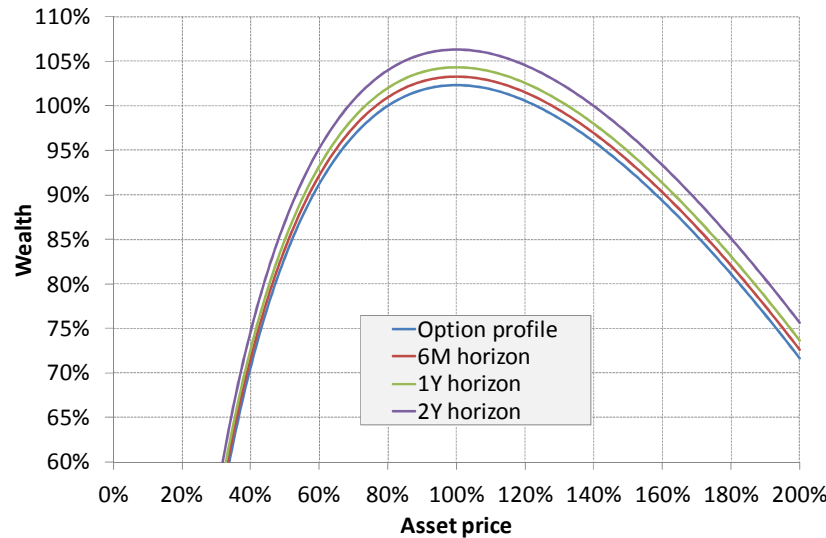
$f(S_t) \times S_t$ is the total amount of the risky asset bought at time t and m is a scaling coefficient. This investment rule is illustrated in Figure 8. Obviously, this investor has a short position when the asset price is above the average, and a long one if the asset price is below the average. The number of risky asset shares in the portfolio decreases with respect to the asset price. f is effectively a decreasing function of S , as shown in Figure 8. The investor thus takes advantage of volatility when the price oscillates around a given level (even if this level is not the target level S_{target}).

In terms of risks, this exposure policy would be unlimited if the asset price rose to infinity. Moreover, the number of asset shares is unlimited when the asset price goes to 0. Note that this framework makes it possible to set limits on the amount and/or number of shares, by using a different definition of f . This would lead to more acceptable maximum risks for this strategy. However, to keep things simple, this is not done here. In this situation, proposition 1 applies straightforwardly. The option profile and the trading impact are equal to:

$$\begin{aligned} \text{Option profile} &= m \times (S_{target} \ln(S_T) - S_T) \\ \text{Trading impact} &= \frac{1}{2} m S_{target} \int_0^T \sigma_t^2 dt \end{aligned}$$

As expected, the trading impact is always positive and proportional to the realised variance of the asset during the investment period. Thus, for a given final value of the asset, the final wealth increases with the realised variance. Conversely, the option profile is concave and can potentially lead to unlimited losses.

Figure 9: Investor wealth ($S_0 = 80\%$, $S_{target} = 100\%$)



4.1.2 Analysis of the option profile

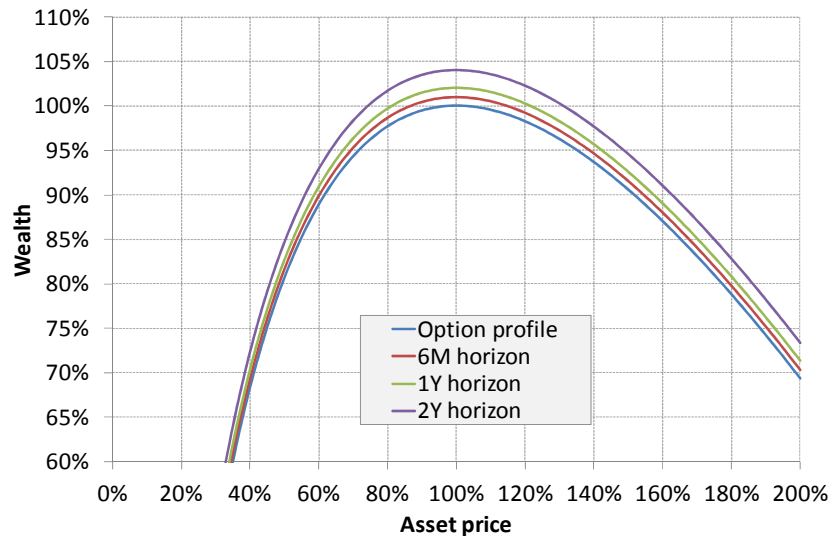
Suppose that the asset price at initial date is not equal to the average price. A significant profit can be made if the asset price is closer to the average at maturity than it was at inception date. Figure 9 shows the final wealth of an investor as a percentage of the initial wealth. Suppose that the initial price of the asset is equal to 80% of the long term average $S_{target} = 100\%$. Over a six-month time horizon, a significant mid-term profit can be generated if the asset price moves closer to the average. If the price increases from 80% to 100%, the realised gain is 3.3%. Conversely, significant losses are incurred if the asset price decreases. If the price falls from 80% to 60%, realised losses are equal to 7.7%. Nevertheless, investors may accept this risk if they strongly believe that the asset price will converge to S_{target} in the near future.

4.1.3 Trading Impact

Now, suppose that the investor starts with initial wealth $X_0 = 100\%$. Suppose also that the initial price S_0 of the asset is equal to the average price S_{target} . Figure 10 shows the wealth of the investor at time T as a function of S_T . We also assume that the annualised volatility of the risky asset is a constant 20%. In this figure, the option profile is always lower than the initial wealth of the investor. Mathematically, $F(S_T)$ is negative for all S_T . The option profile always makes a negative contribution to the performance. Naturally, this loss increases when the asset price moves away from the average. Profits come from the trading gains. These gains increase when the strategy is performed for a longer time.

This strategy thus delivers performance when, after a volatile trajectory, the asset price finishes near the average. On the other hand, significant short-term losses can occur if the price moves away from the average. This is therefore an asymmetric strategy, involving small and slow gains with high probability and large and quick losses with low probability.

Figure 10: Investor wealth ($S_0 = S_{target} = 100\%$)



4.2 Averaging down

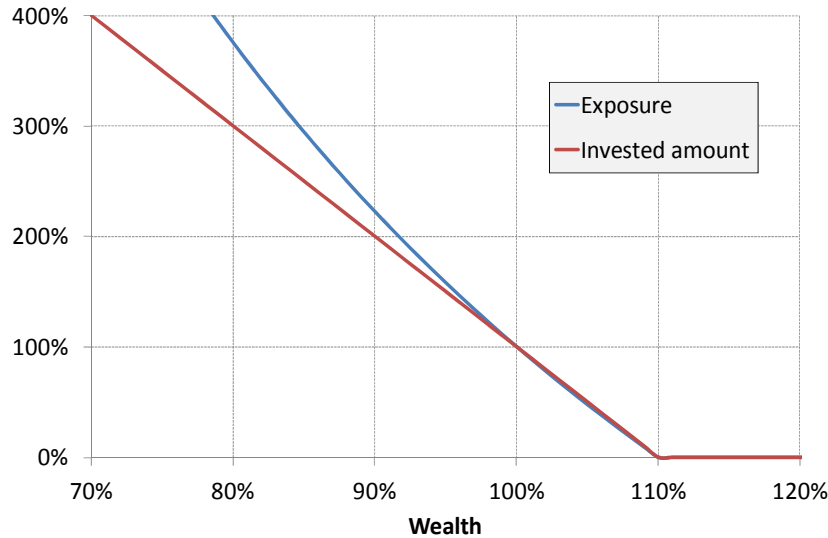
In this section, we will show that this type of strategy is highly likely to deliver gains, balanced by a significant bankruptcy risk.

4.2.1 A miracle recipe for recovering losses?

Let us start with an example. Suppose that an investor buys a stock at \$100 price, and that the stock price drops to \$90. The difference between the average entry price and the current price is \$10. If the investor buys another share in the same stock (i.e. doubles his position), the average buying price is now \$95. The difference between the current price and the average entry price is now only \$5. Of course this new figure does not correspond to any actual loss reduction. The \$10 losses are just diluted into a larger position. As the exposure is larger, a small \$5 increase of the stock price is now sufficient to cancel out all previous losses. But the larger exposure to the asset may also exacerbate future price falls. A \$10 asset price decrease will now lead to a \$20 loss. Some investors may be tempted to average down once again, in order to take advantage of a potential rebound. On the other hand, the investor may face severe risks after doubling his exposure a few times.

This strategy can be related to the martingale gambling technique. It was originally designed for a game in which a gambler doubles his stake if his bet is successful, or else loses it. The martingale strategy is to double the bet after every loss, so that the first win would recover all previous losses, plus a profit equal to the original stake. The gambler will almost surely win if he is allowed to bet an infinite number of times (see Harrison and Kreps (1979) for a detailed analysis). Unfortunately, he may go bust before his first win in real life situations, as his stakes double at each loss. Indeed, after 5 consecutive losses, the gambler has to bet 32 times his original stake, which is unacceptable in most real life situations. The losses in the most adverse scenario (going bust) are of several orders of magnitude over the expected gains. For the same reasons, averaging down strategies can recover losses if a small price increase occurs, but may result in the investor going bust if this increase happens too late.

Figure 11: Exposure policy as a function of the wealth level



Clearly, averaging down could absorb losses provided the asset price stays above a certain limit. Beyond this limit, the strategy may lead to huge losses due to the increasing exposure to the asset. The goal of this section is to identify this threshold.

4.2.2 A fine line between objective achievement and severe losses

Let us formalise this strategy. As previously, X_t stands for the investor's wealth at time t , and S_t for the price of the risky asset. Suppose that an investor has an initial wealth $X_0 = 100\%$, and wants to obtain a target wealth of $X_{target} = 110\%$ whenever the asset price increases by $R = 10\%$. Initially, this objective can be attained by investing 100% of the wealth in the risky asset. Now, suppose this investor starts from a wealth level of X_t . The exposure needed to obtain target wealth of X_{target} if the underlying asset moves by 10% is described by the following relationship:

$$X_{target} = X_t (1 + R \times e(X_t))$$

or equivalently:

$$e(X_t) = \frac{1}{R} (X_{target} - X_t) \quad (6)$$

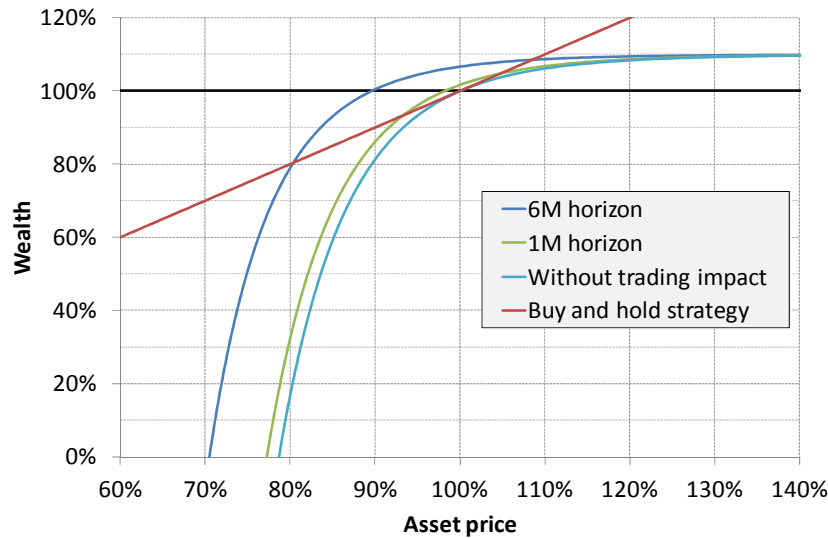
The variation of wealth will be governed by equation (3) but with variable exposure. The determining factor in this strategy is the distance between current wealth and targeted wealth. It shares some similarities with contrarian strategies in which the determining factor is the distance between the current asset value and a target asset value. This exposure policy is decreasing with respect to current wealth, as shown in Figure 11. Intuitively, it is an increasing function of the objective, as more risks must be taken to achieve higher objectives.

Proposition 3 Consider a portfolio strategy that follows the exposure rule (6). The distance to the target wealth can be broken down into an option profile and some exponential trading impact:

$$X_{target} - X_T = \underbrace{(X_{target} - X_0) \left(\frac{S_T}{S_0} \right)^{-\frac{1}{R}}}_{\text{option profile}} \exp \left(\underbrace{-\frac{1}{2} \left(\frac{1}{R} + \frac{1}{R^2} \right) \int_0^T \sigma_t^2 dt}_{\text{trading impact}} \right) \quad (7)$$

First of all, the objective is only attained asymptotically, as exposure converges to 0 as the strategy approaches the objective. The option profile is always below the static strategy with similar initial exposure, as shown on Figure 12. The averaging down overperformance may only come from trading impact (i.e. volatility). The longer the investment period, the higher this trading impact will be. In this example, volatility is presumed to be constant and equal to 20%.

Figure 12: Averaging down strategy outcome

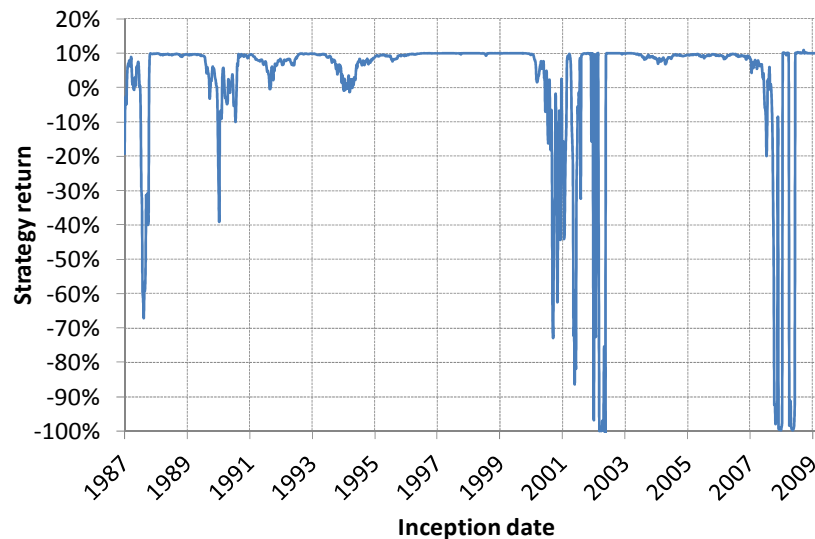


While the strategy profile stays close to the objective when performance is positive, it decreases very quickly for negative returns. In this example, the profile leads to bankruptcy when the asset price decreases by 20%. Indeed, the risky asset return objective R is typically low compared to 100%. Therefore, the term $\left(\frac{S_T}{S_0} \right)^{-1/R}$ may be a very large negative power of the spot price⁵. This large power explains the sudden loss behaviour of the option profile.

From a financial point of view, this is explained by the increasing exposure when the strategy moves away from the objective. Over a six month horizon, the strategy delivers a positive P&L as soon as the risky asset has a return greater than -10% . If asset volatility is equal to 20%, the probability of a positive P&L after 6 months is 76%. Unfortunately,

⁵In Figure 11, we take $R = 10\%$. Therefore the option profile involves a power $-\frac{1}{R} = -10$ of the underlying asset.

Figure 13: 1Y return of the average down strategy



this attractive hit ratio is counter-balanced by potentially very large losses. The probability of going broke within six months is 2%. This is low, but not negligible, as this single event could absorb years of profits plus the initial capital.

4.2.3 Backtesting

This asymmetry can be shown in a backtest of this strategy on the DJ Eurostoxx 50 index. We provide a series of one-year simulations, with starting dates ranging from January 1987 to December 2010. The return objective is set at 10%. The resulting one-year returns are displayed in Figure 13. One-year returns meet the objective for long time periods. For example, these returns are very stable from 1991 to 2000, providing comforting results during those nearly ten years. On the other hand, it leads to severe collapses, or even going broke during the market downturns of 2002 and 2008.

5 Trend-following strategies

Trend-following strategies are a specific example of an investment style that emerged as an industry in its own right. So-called Commodity Trading Advisors are the largest sector of the Hedge Fund industry. According to BarclayHedge⁶, the total AUM managed by CTA as of Q1 2011 was \$290Bn AUM, equivalent to 15% of total Hedge Fund AUMs at that date.

This said, trend-following styles are not restricted to CTA funds, as they have been used by many other investment managers for a long time. They were mentioned, for instance, in Graham (1949):

“In this respect the famous Dow Theory⁷ for timing purchases and sales has had an unusual history. Briefly, this technique takes its signal to buy from a special kind of break-through of the stock averages on the up side, and its

⁶See www.barclayhedge.com/research/indices/cta/Money_Under_Management.html.

⁷An outmoded denomination of momentum or trend-following techniques.

selling signal from a similar break-through on the down side. The calculated-not necessarily actual-results of using this method show an unbroken series of profits in operations from 1897 to 1946. On the basis of this presentation the practical value of the Dow Theory would appear firmly established; the doubt, if any, would apply to the dependability of this published record as a picture of what a Dow theorist would actually have done in the market.”

Surprisingly, despite its importance in the investment industry, this investment style is largely overlooked by standard finance textbooks. Most available documents about trend-following techniques consist of a collection of testimonials of how successful traders managed to make money out of their trading rules. It is difficult to identify a clear direction among those publications. Of these, let us quote Turtles (2003), which offers a valuable introduction to CTA systems⁸. This text summarises the trading principles advocated by Richard Dennis, a pioneer of systematic CTA trading.

From a very different perspective, as mentioned in the introduction, Fung and Hsieh (2001) propose to benchmark trend-following strategies against the returns of a lookback straddle on the MSCI World. They follow a line of thought similar to Proposition 1 and infer this profile from the qualitative properties of the investment process.

This section is devoted to providing a simplified analysis of those strategies. While diversification across a large number of assets greatly improves the efficiency of trend-following strategies, to keep things simple we are going to focus on the single asset case here. First, trend-following strategies are analysed within a simple tree model, which identifies the qualitative properties of CTA strategies. In a second section, a more refined model is proposed, which gives a precise representation of trend-following strategies in terms of option profile and trading impact.

5.1 Analysis of trend-following strategies in a toy model

This section provides stylised facts about trend-following strategies in a binomial tree model. It is borrowed from the first part of Potters and Bouchaud (2005). In this model, the risky asset starts with an initial price of \$100. This price increases or decreases by \$1 at each period, with a similar probability. The period would typically be one day. Figure 14 describes the cumulated P&L of a corresponding long investment in this asset for the next three periods.

In this framework, the simplest trend-following strategy would be the following. Suppose that an investor observed a positive trend before the start date. His strategy is to invest \$100 in the asset, and to sell this position after the first negative return is observed (which can be interpreted as a negative trend). Figure 15 represents the P&L of this strategy depending on the risky asset trajectory. We also compute the probability of each outcome, assuming that the positive and negative return probabilities are both equal to 50%. The final P&L is represented in light blue boxes.

In this simple model, the loss probability is 50%, while the gain probability is only 25% (the remaining 25% corresponding to neither profits or losses). However, while losses are limited to \$1, the gains can go up to any value. The average gain is \$2, which offsets the smaller gain probability. Thus, the returns distribution of this strategy is positively skewed: small losses are frequent, while gains occur rarely, but with a larger amplitude. This is

⁸The text “The original Turtle Trading Rules” may be found on the web site www.originalturtles.org.

Figure 14: P&L of a risky asset investment in the binomial model

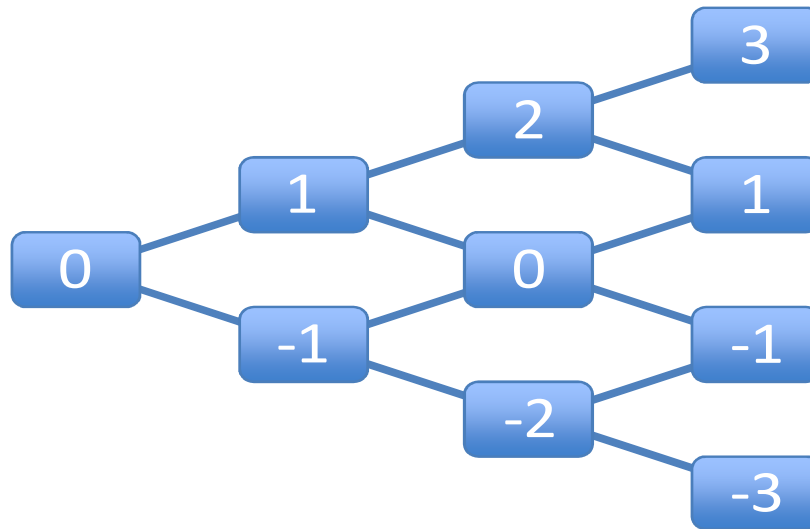


Figure 15: P&L of the trend-following strategy

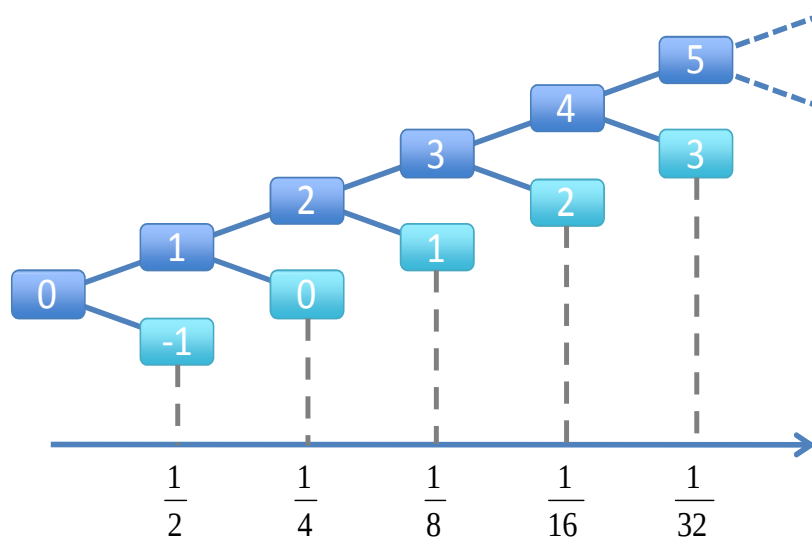
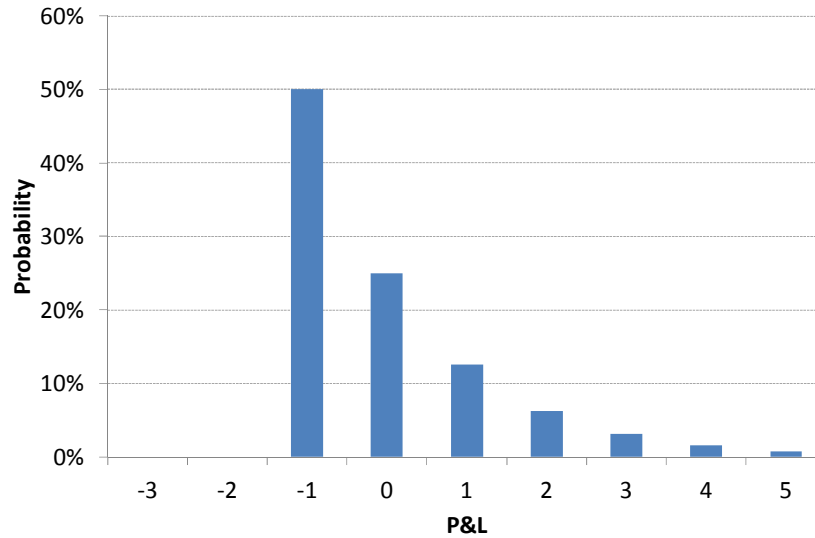


Figure 16: P&L distribution of the trend-following strategy



illustrated in Figure 16. With Proposition 1 in mind, this is very consistent with a strategy exhibiting a convex option profile and negative trading impact. Let us now be more precise and obtain the corresponding breakdown in a continuous time model.

5.2 Trend-following strategies as functions of the observed trend

5.2.1 A model of a trend-following strategy

Trend-following estimators use past returns to predict future price changes. In the previous example, the trend was estimated using only the last return: the investor forecasts a positive (resp. negative) return if the last one was positive (resp. negative). In reality, longer period returns, such as one month or one year past returns, are used for estimation. Many trend followers use moving averages of prices or returns in order to provide more stable predictions. For example, some consider the difference between a short-term (e.g. one month) moving average of the asset price, and a long-term average (e.g. six month). This difference is positive when past returns are mostly positive during the last six month period. It is therefore considered to be a valid estimator of past trends. In real life situations, investors use more complex combinations of such indicators.

Our purpose here is not to find the best estimate, but to provide clear analysis. We will therefore consider the simple example of a moving average of past daily returns. We choose a moving average with exponential weights:

$$\hat{\mu}_t = \frac{1}{\tau} \sum_{i=0}^n e^{-\frac{i\delta t}{\tau}} \frac{S_{t-i\delta t} - S_{t-(i+1)\delta t}}{S_{t-(i+1)\delta t}}$$

Such an estimator has interesting properties. It depends on a single parameter τ , which represents the average duration of estimation. Due to exponential weighting, recent returns have a larger impact than older ones. It does not suffer from “threshold effect”, that is, changes of regimes due to a past observation that exits the averaging window. Moreover, this estimator has theoretical foundations, as it can be interpreted as the Kalman filter for an

unobservable trend. Lastly, it produces simple formulas for the performance of the related strategy. In particular, one can derive the dynamics of the moving average depending on the asset returns:

$$\hat{\mu}_{t+\delta t} = \left(1 - \frac{\delta t}{\tau}\right) \hat{\mu}_t + \frac{1}{\tau} \frac{S_{t+\delta t} - S_t}{S_t}$$

or, in a continuous time framework:

$$d\hat{\mu}_t = -\frac{1}{\tau} \hat{\mu}_t dt + \frac{1}{\tau} \frac{dS_t}{S_t} \quad (8)$$

In the following, we suppose that the investor considers that the best returns estimation between times t and $t + dt$ is $\hat{\mu}_t dt$. Based on this assumption, the investor can simply apply the optimal Markowitz/Merton strategy⁹. Here, we will still consider that the risk-free rate is 0. In this case, its exposure to the risky asset will be:

$$e_t = m \frac{\hat{\mu}_t}{\sigma^2} \quad (9)$$

where μ is the trend estimator, σ is the annualised volatility of the underlying, and m is a risk tolerance parameter. Interestingly, this exposure strategy is qualitatively consistent with the principles described in the Turtles Rules. The exposure to the risky asset is proportional to this risk tolerance. This parameter depends on the investor's risk profile. This exposure is also proportional to the trend estimator value $\hat{\mu}_t$ and inversely proportional to the risk. Note that it is not capped and is almost never 0.

Following this strategy, the investor's wealth evolves as equation (3):

$$\begin{aligned} \frac{dX_t}{X_t} &= e_t \frac{dS_t}{S_t} \\ &= m \frac{\hat{\mu}_t}{\sigma^2} \frac{dS_t}{S_t} \end{aligned} \quad (10)$$

Now, by inverting equation (8), the dynamic of the wealth can be written as a function of the trend $\hat{\mu}_t$. We can then follow the steps of Proposition 2 where the variable is no longer the asset price S_t but the asset trend $\hat{\mu}_t$.

Proposition 4 *The logarithmic return of a trend-following strategy that follows the exposure function (9) can be broken down into an option profile and some trading impact:*

$$\ln \frac{X_T}{X_0} = m \left(\underbrace{\frac{\tau}{2\sigma^2} (\hat{\mu}_T^2 - \hat{\mu}_0^2)}_{\text{option profile}} + \underbrace{\int_0^T \left[\frac{\hat{\mu}_t^2}{\sigma^2} \left(1 - \frac{1}{2}m\right) - \frac{1}{2\tau} \right] dt}_{\text{trading impact}} \right) \quad (11)$$

It is worth remembering that the trend $\hat{\mu}_t$ is not a model assumption but the actual measured trend.

5.2.2 Option profile

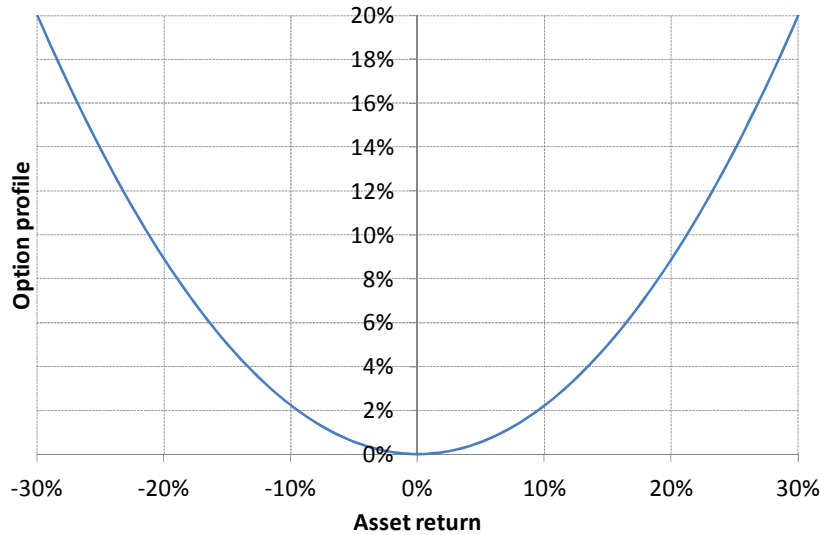
The option profile is related to the square of the observed trend:

$$\text{Option profile} = m \frac{\tau}{2\sigma^2} (\hat{\mu}_T^2 - \hat{\mu}_0^2)$$

⁹See Merton (1971) for details.

This is similar in concept to the straddle profile suggested by Fung and Hsieh (2001). It is also perfectly in line with regressions including squared returns factors performed by Harvey and Siddique (2000). This option profile part evolves quickly, and has a short term memory. It vanishes along with the measured trend $\hat{\mu}$. Interestingly, the worst case scenario is that the measured trend falls from its initial value $\hat{\mu}_0$ to 0, which would lead to an upper limit of $\frac{m\tau}{2\sigma^2} \hat{\mu}_0^2$. This quantity depends largely on the value of the measured trend at the start. This is natural, as the strategy is more exposed to the risky asset when the trend is high. This option profile is displayed in Figure 17, where we consider a six month moving average, with 15% risky asset volatility, and a risk tolerance parameter $m = 5\%$ (see the next section for details about this set of parameters). Furthermore, we assume that the initial measured trend is equal to $\hat{\mu}_0 = 0$. As a consequence, the option profile is always positive.

Figure 17: Short term option profile, as a function of the risky asset return



5.2.3 Trading impact

On the other hand, trading impact is obtained by a cumulated function of the realised trend:

$$\text{Trading impact} = \int_0^T m \left[\frac{\hat{\mu}_t^2}{\sigma^2} \left(1 - \frac{1}{2}m \right) - \frac{1}{2\tau} \right] dt \quad (12)$$

This trading impact changes slowly, and has a long-run memory due to its cumulative nature. The evolution of this long term P&L depends only on the ratio $\frac{\hat{\mu}_t^2}{\sigma^2}$ which is the square of the measured short term Sharpe ratio. Trading impact is negative on the long term if this ratio is below $\frac{1}{\tau(2-m)}$, and otherwise positive.

The risk tolerance m can be related to the target volatility σ_{target} by setting $m = \sigma_{target} \sqrt{2\tau}$. In practical examples, m (around 10%) will be small with respect to 1. Thus, a good estimate for long term trading impact per annum is given by $\frac{\hat{\mu}_t^2}{\sigma^2} - \frac{1}{2\tau}$. This provides a necessary condition for the trend-following strategy to generate profits:

$$|\text{Sharpe ratio}| > \frac{1}{\sqrt{2\tau}}$$

For a 6 month moving average (i.e. $\tau = \frac{1}{2}$), the long-term profit increases if the annualised measured Sharpe ratio is above one in absolute terms.

Eventually, the worst-case scenario for the long term P&L is that $\hat{\mu}_t = 0$, which is no surprise! In that case, annual losses due to trading impact are capped at $\frac{m}{2\tau} = \frac{\sigma_{target}}{\sqrt{2\tau}}$. In the case of a six-month moving average, annual maximum long-term losses are equal to the target volatility of the strategy.

5.2.4 Backtested example

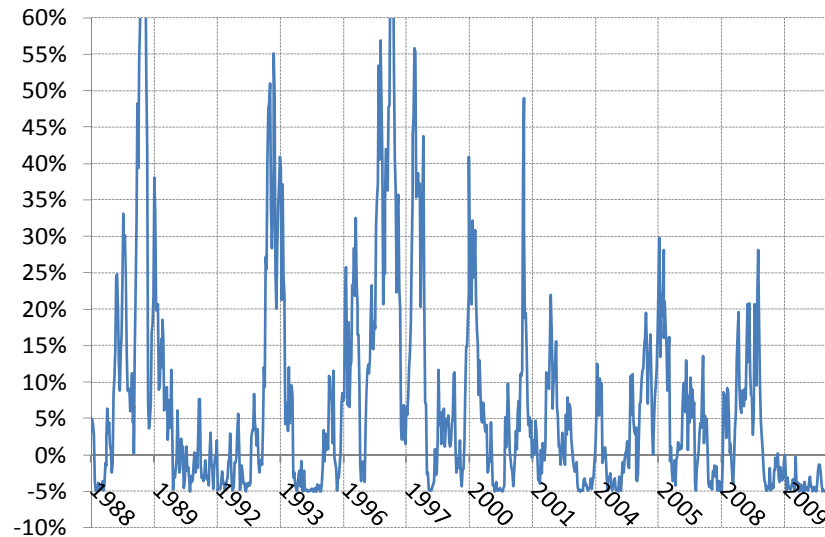
In this section, the trend-following strategy is backtested on the DJ Eurostoxx 50 index. Instead of performing the trend-following strategy directly on the index, it is run on a volatility targeted index which constantly adjusts its exposure to the Eurostoxx 50 in order to keep its volatility equal to $\sigma = 15\%$. The strategy itself uses a six-month moving average ($\tau = \frac{1}{2}$), with target volatility of $\sigma_{target} = 5\%$. This involves a parameter $m = 5\%$, with an average absolute exposure above 33% (the average backtested absolute exposure is around 50%).

Figure 18: Backtest of a trend-following strategy on the vol-target Eurostoxx 50



The backtest is performed between January 1987 and May 2011. Figure 18 shows the breakdown of the strategy value between the option profile and the trading impact. Long term trading impact is computed from formula (12), independently of the strategy value calculations. This component is smoother than the strategy NAV. The short-term option profile is the difference between the strategy NAV and the long term P&L. This component is always positive, as the measured trend at the start is set at 0. The strategy NAV is therefore always above the long term value. By construction, this long term cannot decrease more than 5% p.a. (the instantaneous slope of the long term P&L, i.e. the daily trading impact, is given in Figure 19). The worst-case scenario thus appears fairly acceptable.

Figure 19: Annualised daily contribution to trading impact



5.3 Asymmetrical return distribution

The return distribution of the strategy depends heavily on the distribution on the trend estimator $\hat{\mu}$, as shown in equation (11). Of course, the behaviour of $\hat{\mu}$ depends of the model assumptions on the asset price dynamics. Let us suppose a standard Gaussian dynamic for S_t with fixed volatility σ and a fixed trend μ . In this case, the long term distribution of the trend estimation is Gaussian, with an average of μ and volatility of $\frac{\sigma}{\sqrt{2\tau}}$.

In the case of the six-month exponential moving average, the standard deviation of the measured trend is equal to the standard deviation of the yearly return of the underlying asset. The long term returns of the strategy are driven by the square of the measured trend μ^2 and are then highly asymmetric. The stationary distribution of the annualised contribution to the long term P&L is given in Figure 20, in the same conditions as in the backtest of the previous section. This distribution is computed under two sets of assumptions: in the first one, the underlying asset has no drift (risk neutral probability), while in the second one the Sharpe ratio is constant and equal to 1 (this involves a constant drift $\mu = 15\%$).

This distribution looks very similar to the toy model version of Figure 16. The general result remains unchanged. The probability of losing is higher than the probability of gaining, but the average gain is much higher than the average loss. As expected, the trend-following strategy works poorly in trendless models (the average P&L is equal to 0), and much better in models with a positive trend. For an easier comparison of the winning and losing probabilities, the cumulative version of the distribution is illustrated in Figure 21. When there is no trend, the positive P&L probability is around 30% as opposed to 50% in the case where the Sharpe ratio is equal to 1.

Figure 20: Stationary distribution of the annualised daily trading impact

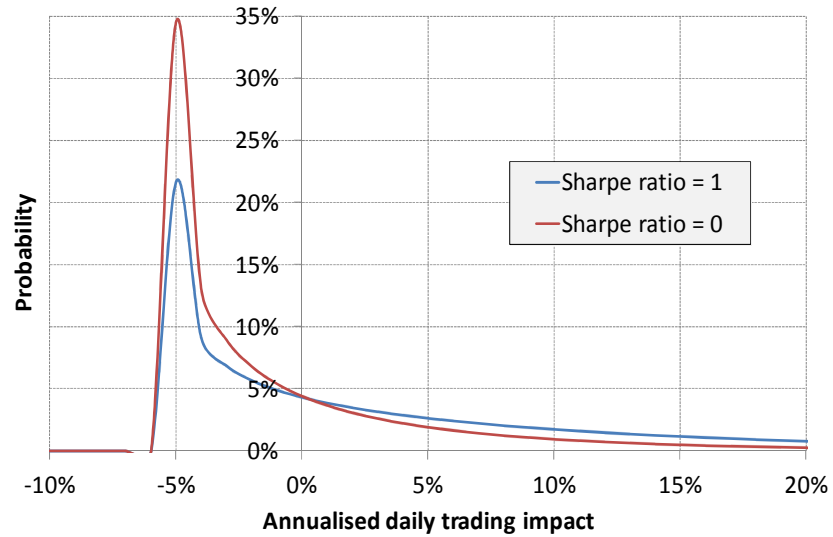
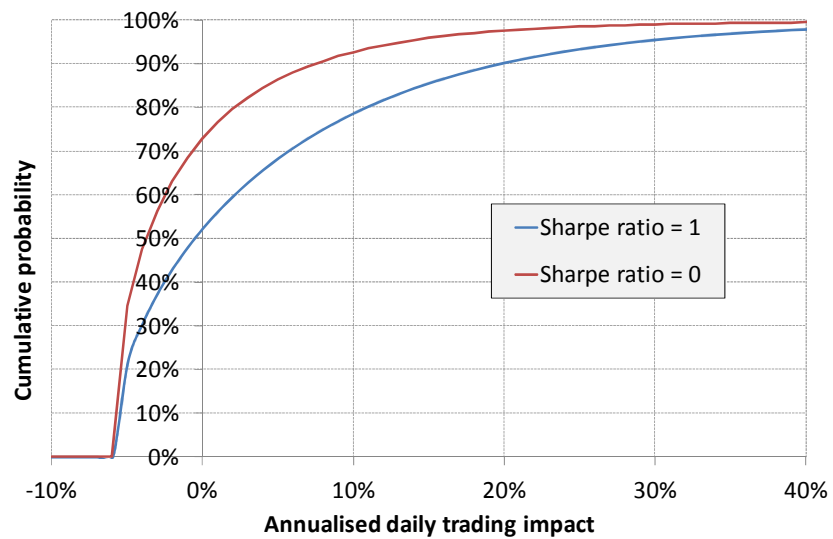


Figure 21: Cumulative distribution of the annualised daily trading impact



6 Concluding remarks

In this paper, we have shown how the returns on many popular strategies can be broken down into an option profile and some trading impact, both being of similar importance, on average. We reviewed three classes of strategies (directional, contrarian and trend-following), and obtained a quantitative breakdown for each of these. This breakdown allows for an accurate risk-return analysis of each of those strategies. More specifically, it illustrates how the probabilities of making gains or losses have to be analysed together with the corresponding average amount of gain or losses. High hit ratios are not necessarily signs of good strategies, in fact they can reveal exposure to extreme risks.

Analysis of real-life situations would involve extending the single asset scenario to multi-asset situations, for a better understanding of the impact of adding such strategies. It would also be interesting to build econometric tests to assess whether our model can provide accurate predictions of the behaviour of specific hedge fund strategies. This has been left for further research.

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